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An indexed set representation based multi-objective evolutionary approach for mining diversified top-k high utility patterns

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ABSTRACT

How to discover top-k patterns with the largest utility values, namely, mining top-k high utility patterns, is a hot topic in data mining. However, most of the existing works for mining top-k high utility patterns consider each pattern separately during the mining process, thus many mined patterns are highly similar and lack diversity. In this paper, we propose to mine top-k high utility patterns with high diversity for enhancing users' satisfaction in recommendation. Specifically, we first introduce a simple measure of *coverage* to quantify the diversity of the whole set, that is, the top-k patterns as a complete entity. Then we propose an **i**ndexed **s**et **r**epresentation based **m**ulti-**o**bjective **e**volutionary **a**pproach named ISR-MOEA to mine diversified top-k high utility patterns, due to the fact that the two measures utility and coverage are conflicting. In ISR-MOEA, an indexed set individual representation scheme is suggested for fast encoding and decoding the top-k pattern set. Experimental results on six real-world and two synthetic datasets demonstrate the effectiveness of the proposed approach. The proposed approach can obtain several groups of top-k pattern set with different trade-offs between utility and diversity in only one run, which would further enhance the satisfaction of users.

1. Introduction

Top-k high utility pattern mining, namely discovering k patterns with the largest utility value (or profit) from transaction database, has recently attracted a lot of research work in data mining area (Wu et al., 2012; Yin et al., 2013; Zihayat and An, 2014; Ryang and Yun, 2015; Tseng et al., 2016). In the task of utility mining, each item is associated with a utility (e.g. unit profit) and can appear many times (e.g. quantity) in different transactions. The importance of a pattern can be measured by its utility in terms of weight, value or other information specified by users. Most of the existing works for mining top-k high utility patterns focus on improving the efficiency of the mining algorithms and the mined patterns are considered separately during the mining process. In other words, the item difference between any two patterns is not considered at all in these proposed methods. Thus, the recommended top-k patterns may be very similar and lack diversity.

Take the database shown in Table 1 as an example, if the measure utility is only considered, then the returned top-2 high utility patterns are $\{Jacket, Shoe, Belt\}$ and $\{Jacket, Shoe, Hat\}$. It can be found that there exist many similar items such as $Jacket$ and $Shoe$ among the 2 recommended patterns. In other words, the recommendation of the top-2 high utility patterns lacks diversity, which undermines users' satisfaction. In fact, the following 2 patterns $\{Jacket, Shoe, Belt\}$, $\{Watch, Suitcase, Scarf\}$ may be much better than the top-2 high utility patterns since a practical decision maker system should not only make high utility but also diversified pattern recommendations to improve overall satisfaction of users (Hammar et al., 2013; Wu et al., 2016). Fig. 1 gives two such examples for goods recommendation, where the recommendation-2 by considering both utility and diversity is much better than the recommendation-1 by only considering utility.

To this end, this paper proposes to mine both diversified and high utility patterns for recommendation. In this problem, the recommended top-k patterns are considered as a complete entity, denoted as S. In addition to using utility to measure the profits of S, we introduce a simple measure of coverage to quantify the diversity of S. By considering both utility and coverage, we can obtain both high utility and diversified patterns. However, the measure utility and coverage conflict with each other. In other words, a higher utility of S will lead to lower coverage, whereas a lower utility of S often leads to higher coverage. Due to

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Fig. 1. The examples of two recommended top-2 patterns.

the conflicting property between utility and coverage, an indexed set representation based multi-objective evolutionary approach termed ISR-MOEA is then proposed to obtain diversified top-k high utility patterns. In summary, the main contributions of this paper are described as follows:

- A novel task in utility mining named *mining diversified top-k high utility patterns* is proposed, motivated by the fact that the recommended top-k patterns should not only have high utility but also be diversified. In the suggested task, a measure of coverage is firstly introduced to quantify the diversity of top-k patterns. Then, this task can be formulated as a 2-objective optimization problem since that the two measures utility and coverage are conflicting. To the best of our knowledge, this is the first work by incorporating diversity into top-k high utility mining.
- To tackle this new problem, an indexed set representation based multi-objective evolutionary algorithm named ISR-MOEA is then proposed for mining diversified top-k high utility patterns. In ISR-MOEA, a simple yet effective population initialization strategy is proposed to guarantee that all generated solutions are useful and hold a good diversity for recommendation. In addition, two effective evolutionary operators (i.e. crossover and mutation) are proposed to speed up the convergence of populations.
- The effectiveness and efficiency of the proposed algorithm ISR-MOEA are verified on six real world and two synthetic datasets with different characteristics. Experimental results show the superior performance of our method over several state-of-the-art baseline methods in mining diversified top-k high utility patterns, which indicates that the proposed ISR-MOEA is a competitive and promising method for the task of mining diversified top-k high utility patterns.

The rest of the paper is listed as follows. In Section 2, we introduce the preliminaries about pattern miming, problem formulation, multi-objective evolutionary approach and related work. In Section 3, we give the proposed algorithm in detail. We present our experimental results in Section 4 and conclude the paper in Section 5.

2. Backgrounds and related work

In this section, we first present some preliminaries about utility pattern mining, then give the problem formulations and discuss multi-objective evolutionary approaches. Finally, we give the related work.

Table 1
A toy transaction table for a goods dataset.

t_{id}	Transaction	uti(t_i)
t_1	$\{Jacket[12], Shoe[15], Belt[30], Hat[24]\}$	81
t_2	$\{Jacket[14], Shoe[15], Belt[26], Watch[12]\}$	67
t_3	$\{Jacket[12], Shoe[9], Hat[20], Watch[16]\}$	57
t_4	$\{Jacket[8], Hat[4], Suitcase[15]\}$	27
t_5	$\{Watch[24], Suitcase[27], Scarf[32]\}$	83

2.1. Preliminaries about utility pattern mining

Let transaction database $\mathcal{D} = \{t_1, t_2, \dots, t_m\}$ be a set of transactions, where each transaction t_{id} is a set of items associating with weights (e.g., profits obtained by selling the item). Suppose there are n distinct items in $\mathcal{D}$, denoted as $\mathcal{I} = \{i_1, i_2, \dots, i_n\}$. A *pattern*, denoted as P, is a non-empty set of items. The transactions that contain a pattern P are the *supporting transactions* of P, which is denoted as $\mathcal{D}_P$. The *support* of a pattern P, denoted as $sup(P)$, is the percentage of transactions in $\mathcal{D}$ that contains P. P is called a *frequent pattern*, if $sup(P)$ is no less than a user-specified minimum support threshold *min_sup* ($0 \le min_sup \le 1$).

The utility of one item i in one transaction t is denoted as $uti(i,t)$. The *utility of a pattern* P in one transaction t, denoted as $uti(P,t)$, is the sum of the utilities of P in t, i.e., $uti(P,t) = \sum_{i \in P} uti(i,t)$. The *utility of a pattern* P, denoted as $uti(P)$, is the sum of the utilities of P in its all supporting transactions, i.e., $uti(P) = \sum_{i \in P \subseteq t, t \in \mathcal{D}_P} uti(i,t)$. The *utility* of a database $\mathcal{D}$, denoted as $uti(\mathcal{D})$, is the sum of the utilities of distinct items in its all supporting transactions, i.e., $uti(\mathcal{D}) = \sum_{i \in t, t \in \mathcal{D}} uti(i,t)$. The *relative utility* of a pattern P, denoted as $ruti(P)$, is the fraction of $uti(P)$ to $uti(\mathcal{D})$, i.e., $ruti(P) = uti(P)/uti(\mathcal{D})$. A pattern P is called *high utility pattern* if $ruti(P)$ is not smaller than a user-specified minimum utility threshold *min_uti* ($0 \le min_uti \le 1$). The above definitions about frequent and utility pattern mining can be referred in Zhang et al. (2016) and Tseng et al. (2016).

Table 1 gives one example of a transaction database with utilities, where there are 7 unique items *{Jacket, Shoe, Belt, Hat, Watch, Suitcase, Scarf}* and 5 transactions $\{t_1, t_2, t_3, t_4, t_5\}$. Suppose $P = \{Jacket, Shoe, Belt\}$, the supporting transaction database of P is $\mathcal{D}_P = \{t_1, t_2\}$ and $sup(P) = 2/5$. Suppose *min_sup* = 0.30, P is a frequent pattern. The utility of $Jacket$ in transaction t_1 is $uti(Jacket, t_1) = 12$. The utility of P in transaction t_1 is $uti(P, t_1) = 12 + 15 + 30 = 57$, the utility of P in transaction t_2 is $uti(P, t_2) = 14 + 15 + 26 = 55$, thus $uti(P) = 57 + 55 = 112$. The utility of $\mathcal{D}$ is $uti(\mathcal{D}) = 315$. The relative utility of P is $ruti(P) = 112/315 \approx 0.36$. Suppose *min_uti*= 0.30, P is a high utility pattern.

Based on the definition of utility for a single pattern, we can easily extend the utility definition for a pattern set, specifically,

Definition 1. The *relative utility* of the pattern set S containing k patterns is defined as

$$ruti(S) = \frac{1}{k} \sum_{P \in S} ruti(P).$$

Mining top-k high utility patterns: Given a transaction database $\mathcal{D}$, the task of *mining top-k high utility patterns* is to find the pattern set S (containing k patterns) with the largest relative utility value.

In the running example, suppose k=2, the task of *mining top-2 high utility patterns* is to find 2-pattern set S: {$\{Jacket, Shoe, Belt\}$:112, $\{Jacket, Shoe, Hat\}$:92}. From this example, it can be found that the task of mining *top-k high utility patterns* only considers the utility of each pattern in S, and the diversity of items in the whole pattern set S is not considered. In fact, the recommended top-2 patterns $\{Jacket, Shoe, Belt\}$:112, $\{Watch, Suitcase, Scarf\}$:83 may be more interesting since that this set of patterns is more diversified so as to improve the overall satisfaction of users.

Thus, in this paper, we introduce a measure named *coverage* (Zuo et al., 2015), which can be used to measure the diversity of k-pattern set

$\mathcal{S}$ in top-k high utility pattern mining. The definition of coverage for a pattern set $\mathcal{S}$ is given as follows.

Definition 2. The *coverage* of the pattern set $\mathcal{S}$ with the size of k is defined as

$$cov(\mathcal{S}) = \frac{|S_1 \cup S_2 ... \cup S_k|}{\sum_{i=1}^{k} |S_i|}$$

where the numerator $|S_1 \cup S_2 ... \cup S_k|$ is the number of distinct items appearing in $\mathcal{S}$, and the denominator $\sum_{i=1}^{k} |S_i|$ is the total number of items in $\mathcal{S}$. Obviously, without losing the utility, a higher value of coverage indicates a better recommendation.

In the running example, suppose two pattern sets

$$\mathcal{S}^1 = \{\{Jacket, Shoe, Belt\} : 112, \quad \{Jacket, Shoe, Hat\} : 92\}$$

$$\mathcal{S}^2 = \{\{Jacket, Shoe, Belt\} : 112, \quad \{Watch, Suitcase, Scarf\} : 83\}.$$

The coverage of $\mathcal{S}^1$ and $\mathcal{S}^2$ are $\frac{4}{6}$ and $\frac{6}{6}$, respectively. From this example, we can see that the coverage of $\mathcal{S}^2$ is much larger than that of $\mathcal{S}^1$.

It can be observed that the two measures (i.e., relative utility and coverage) conflict with each other. That is, a higher (relative) utility of a pattern set $\mathcal{S}$ will lead to lower coverage of $\mathcal{S}$ in most cases, whereas a lower utility of $\mathcal{S}$ often leads to higher coverage. Take the database show in Table 1 as an example. Suppose two pattern sets $\mathcal{S}^1$={{Jacket, Shoe, Belt}:112, {Jacket, Shoe, Hat}:92} and $\mathcal{S}^2$={{$Shoe$}:39, {$Watch, Suitcase, Scarf$}:83}. The relative utility $ruti(\mathcal{S}^1) = 102/315$ is high whereas the coverage $cov(\mathcal{S}^1) = 2/3$ is low. Likewise, the relative utility $ruti(\mathcal{S}^2) = 61/315$ is low whereas the coverage $cov(\mathcal{S}^2) = 4/4$ is high. In the following, we use utility to replace relative utility for description convenience.

2.2. Mining diversified top-k high utility patterns

In real applications, the profit of the recommended top-k patterns $\mathcal{S}$, which is measured by the utility, should be as high as possible. In addition, the diversity of $\mathcal{S}$, which is measured by coverage, should also be as high as possible to enhance the satisfaction of users. We name this novel problem as *mining diversified top-k high utility patterns*.

This problem can be formulated as a single optimization problem by weighted sum of the two measures: utility and coverage. Formally,

$$\arg\max_{\mathcal{S},|\mathcal{S}|=k} \{(1-\alpha) \cdot ruti(\mathcal{S}) + \alpha \cdot cov(\mathcal{S})\} \tag{1}$$

where α ($0 \le \alpha \le 1$) is a user defined parameter to capture the relative importance of utility and coverage.

However, it is very difficult to combine these two measures as one measure for optimization, since that the parameter α is hard to be determined by users especially when there is not any prior knowledge in real applications (Zhang et al., 2017).

To this end, in this paper, we formulate the problem of *mining diversified top-k high utility patterns* as a multi-objective optimization problem by considering utility and coverage as two objectives, which is formulated as follows.

$$Maximize \quad \mathbf{F}(\mathcal{S}) = (f_1(\mathcal{S}), f_2(\mathcal{S}))^T,$$

$$where \quad \begin{cases} f_1(\mathcal{S}) = ruti(\mathcal{S}) \\ f_2(\mathcal{S}) = cov(\mathcal{S}). \end{cases} \tag{2}$$

It is worth mentioning that in the above problem formulation, it is not necessary for users to set the parameters α, which brings much convenience to users.

2.3. Multi-objective evolutionary approach

Multi-objective Optimization Problems (MOPs) are commonly found in many real-world applications, e.g. in biology, economics and engineering (Deb, 2015). These optimization problems are characterized by multiple objectives that conflict with each other. Formally, a multi-objective optimization problem can be formulated as follows.

$$max \quad \mathbf{F}(X) = (F_1(X), F_2(X), \dots, F_m(X))^T, \tag{3}$$

where $X = (X_1, X_2, \dots, X_n) \in \mathfrak{R}$ is the decision vector and $\mathfrak{R}$ is the n-dimensional decision space.

Given two decision vectors X_A and X_B, we say X_A dominates X_B (denoted as $X_A \succ X_B$) if $F_i(X_A) \ge F_i(X_B)$ for all $i = 1, 2, \dots, m$, and $\mathbf{F}(X_A) \ne \mathbf{F}(X_B)$. Here we only consider maximized form, since that the minimization problem can be easily transformed into the maximized form. Take the database shown in Table 1 as an example, suppose two pattern sets

$$\mathcal{S}^1 = \{\{Jacket, Shoe, Belt\} : 112, \quad \{Jacket, Shoe, Hat\} : 92\}$$

$$\mathcal{S}^2 = \{\{Jacket, Shoe, Belt\} : 112, \quad \{Jacket, Shoe\} : 77\}.$$

The utility and coverage of $\mathcal{S}^1$ and $\mathcal{S}^2$ are:

$$rutil(\mathcal{S}^1) \approx 0.32, rutil(\mathcal{S}^2) = 0.3$$

$$cov(\mathcal{S}^1) = \frac{4}{6}, cov(\mathcal{S}^2) = \frac{2}{5}.$$

In this example, $\mathcal{S}^1$ dominates $\mathcal{S}^2$ since that $rutil(\mathcal{S}^1) > rutil(\mathcal{S}^2)$ and $cov(\mathcal{S}^1) > cov(\mathcal{S}^2)$.

A decision vector $X \in \mathfrak{R}$ is a Pareto optimal solution (also called non-dominated solution) if we cannot find any $X^* \in \Omega$ satisfying $X^* \succ X$. The Pareto set is actually the set of all Pareto optimal solutions, which is defined as $PS = \{X \in \mathfrak{R} | \nexists X^* \in \mathfrak{R}, X^* \succ X\}$. The projection of the Pareto set into the objective space is denoted as the Pareto front, which is defined as $PF = \{\mathbf{F}(X) | X \in PS\}$.

Multi-objective evolutionary approaches (MOEAs) are used to find a set of non-dominated solutions with the aim to approximate the true Pareto front. MOEA/D (Zhang and Li, 2007) is one of popular frameworks for MOEAs. The main idea of MOEA/D is decomposing a multi-objective optimization problem into a number of different single objective optimization subproblems and then designing a population-based approach to optimize these subproblems simultaneously. MOEA/D can generate a set of evenly distributed solutions as well as have good convergence comparing to other MOEAs such as NSGA-II (Deb et al., 2002) and SPEA-II (Zitzler et al., 2001).

2.4. Related work

Generally, the related work can be classified into the following five categories, i.e., high utility pattern mining, top-k high utility pattern mining, evolutionary utility pattern mining, the concept of coverage on pattern mining and multi-objective evolutionary optimization for data mining.

High utility pattern mining. The problem of high utility patten mining was firstly proposed by Yao et al. (2004), where a new measure *utility* was suggested for measuring the profits or utilities of patterns. A pattern is called a high utility pattern if its utility value is not less than a user specified threshold *min_uti*. After that, many efficient algorithms such as UP-Growth (Tseng et al., 2010) and HUI-Miner (Liu and Qu, 2012) were proposed for high utility pattern mining. However, in these algorithms, the parameter *min_uti* should be specified by users, which is very difficult for users in real applications. To this end, researchers in utility mining are beginning to focus on mining top-k high utility patterns without setting a minimal utility threshold, where k is the desired number of high utility patterns to be mined. It is easy for users to set parameter k.

Top-k high utility pattern mining. The problem of *top-k high utility patten mining* by considering both quantities and profits of items was

first proposed by Wu et al. (2012). After that, Yin et al. proposed an efficient algorithm TUS for mining top-k high utility sequential patterns (Yin et al., 2013). Zihayata and An proposed an efficient algorithm T-HUDS for mining top-k high utility patterns over data streams (Zihayat and An, 2014). Ryang and Yun extended the problem (Wu et al., 2012) and proposed the REPT algorithm (Ryang and Yun, 2015) with four strategies for efficient top-k high utility pattern mining. Recently, Tseng et al. proposed two efficient algorithms named TKU and TKO for mining top-k high utility patterns (Tseng et al., 2016). In summary, all these algorithms were proposed for efficient top-k high utility patterns mining, where only the utility of each pattern in the returned top-k patterns is considered and the item difference between any two patterns is not considered at all, leading to that the recommended top-k patterns are very similar and lack diversity.

Evolutionary utility pattern mining. Different from traditional high utility pattern mining, some evolutionary utility pattern mining algorithms were proposed for mining high utility patterns. For example, Kannimuthu and Premalatha presented the first work to use GA-based algorithm with the ranked mutation to mine high utility patterns (Kannimuthu and Premalatha, 2014). However, it is time-consuming to set the appropriate chromosomes at the first stage in their approach. To this end, other different evolutionary computation-based methods such as particle swarm optimization (PSO) (Lin et al., 2016, 2017) and ant colony system (ACS) (Wu et al., 2017) were used for mining high utility patterns effectively and efficiently. Recently, we proposed a multi-objective evolutionary algorithm named MOEA-FHUI for mining frequent and high utility patterns, where the two measures support and utility are considered as the two objectives for optimization simultaneously (Zhang et al., 2018). Compared with our previous work, the novel task of mining diversified top-k high utility patterns proposed in this paper is more complicated and challenging due to the difficulty of individual representation for a pattern set, which means that the techniques proposed in MOEA-FHUI cannot be applied to the proposed task. To solve the challenges, we suggested an indexed set representation scheme to fast encode and decode the top-k pattern set. Based on this representation, we designed an effective multi-objective evolutionary algorithm for mining diversified top-k high utility patterns, where a novel specific population initialization strategy and novel evolutionary operators were proposed for speeding up the convergence of populations.

The concept of coverage on pattern mining. For the area of pattern mining, Raedt and Zimmermann (2007) developed a constraint-based pattern set mining for global pattern mining, where the *redundancy* was designed to measure the degree to which two patterns overlap in their supporting transactions. In addition, Leeuwen and Knobbe (2012) formalized *cover redundancy* and *joint entropy* to measure the redundancy in subgroup sets for diverse subgroup set discovery. Later, they proposed to discover skylines of subgroup sets in terms of quality and diversity, where *joint entropy* was used to measure the diversity of subgroup sets (Leeuwen and Ukkonen, 2013). In summary, the computation of these diversity measures needs the information about the supporting transactions of patterns and the associated label for each transaction in database. Recently, Yang et al. (2016) proposed to mine diversified temporal subgraph patterns, where the *coverage set* (i.e. the number of different node and time pairs in a set of subgraphs) was used for measuring the diversity of a set of temporal subgraphs. Different from these works, the *coverage* introduced in this paper is used to measure the diversify of recommended k patterns by only considering item difference among the pattern set on the datasets without any labels.

Multi-objective evolutionary data mining. Multi-objective evolutionary optimization is one of the most hot topics in evolutionary computation community, and many interesting algorithms such as Cheng et al. (2015) and Zhang et al. (2015b) have been proposed. Note that many multi-objective evolutionary algorithms (MOEAs) have been proposed for solving other problems in data mining (Mukhopadhyay et al., 2014), e.g., classification (Levesque et al., 2012), clustering (Mukhopadhyay et al., 2015), association rule mining (Bidgoli et al., 2013), feature selection (Venkatadri and Rao, 2010), subgroup discovery (Carmona et al., 2010), task-oriented patterns mining (Zhang et al., 2017) and so on. As far as we know, this work is the first study of incorporating the coverage measure into top-k high utility pattern mining.

3. Proposed approach ISR-MOEA

In this section, we will present the proposed indexed set representation based multi-objective evolutionary approach termed ISR-MOEA for mining diversified top-k high utility patterns, including the proposed indexed set based representation scheme, the overall procedure, the population initialization strategy and evolutionary operators.

3.1. The indexed set representation scheme

In evolutionary pattern mining, there are two common individual representations for a single-pattern. One is the binary individual representation (Khatun et al., 2015; Zhang et al., 2018), in which the length of each individual equals to the number of distinct items in database. This representation is effective for evolutionary pattern mining, while the number of distinct items (i.e. $|\mathcal{I}|$) in database should not be large. The other one is index-based individual representation (Lucas et al., 2017), where each individual of the population is represented by one or more integers and each integer (or index) equals to the position of an item in $\mathcal{I}$. This representation is applicable for high dimension evolutionary pattern mining. Based on these individual representations for a single-pattern, the individual representation for a pattern set can be extended in the following.

It is very straightforward to use binary encoding scheme to encode the pattern set $\mathcal{S}$ which contains k patterns. To be specific, if the item appears in each pattern, then the corresponding bit is set to 1 or 0 for otherwise. For example, suppose k=3 and there are 7 items (i.e. $\mathcal{I} = \{Jacket, Shoe, Belt, Hat, Watch, Suitcase, Scarf\}$) appearing in database shown in Table 1. Let pattern set $\mathcal{S} = \langle P_1, P_2, P_3\rangle = \langle\{Jacket, Belt, Scarf\}, \{Hat, Watch, Suitcase\}, \{Jacket, Watch\}\rangle$. The binary encoding of P_1, P_2 and P_3 are $\langle 1,0,1,0,0,0,1\rangle$, $\langle 0,0,0,1,1,1,0\rangle$, $\langle 1,0,0,0,1,0,0\rangle$, respectively. Thus, the pattern set $\mathcal{S}$ is the combination of the 3 patterns, which can be represented as $\langle 1,0,1,0,0,0,1,\Delta,0,0,0,1,1,1,0,\Delta,1,0,0,0,1,0,0\rangle$ (the identifier 'Δ' is used for separating patterns). It can be observed that the binary encoding length of $\mathcal{S}$ is very large and is equal to $|\mathcal{I}| \times k$, which makes MOEAs hard to converge.

Instead of using binary encoding, an indexed set representation scheme is suggested for directly encoding pattern set $\mathcal{S}$ to address the above problem. Specifically, only the index of item appearing in each pattern is used to represent the pattern. For example, suppose $\mathcal{I} = \{Jacket, Shoe, Belt, Hat, Watch, Suitcase, Scarf\}$, the indices of $Jacket, Belt, Scarf$ are $1, 3, 7$ respectively. Thus, pattern $\{Jacket, Belt, Scarf\}$ can be represented as $\{1, 3, 7\}$. Formally, each pattern set $\mathcal{S}$ can be represented as follows.

Definition 3 (*Indexed Set Representation*)**.** An individual $\mathcal{S}$ containing k patterns (i.e. $\mathcal{S} = \langle P_1, P_2, \ldots, P_k\rangle$) can be represented by the combination of the indices of P_i (denoted as S_i for $1 \le i \le k$) and identifier 'Δ', which is denoted as

$$\mathcal{S} = \langle S_1, \Delta, S_2, \Delta, \ldots, \Delta, S_k\rangle \tag{4}$$

where the identifier 'Δ' is used for separating patterns.

For example, suppose $\mathcal{S} = \langle P_1, P_2, P_3\rangle = \langle\{Jacket, Belt, Scarf\}, \{Hat, Watch, Suitcase\}, \{Jacket, Watch\}\rangle$, then the indexed set representation of $\mathcal{S}$ is $\langle 1,3,7,\Delta,4,5,6,\Delta,1,5\rangle$. It can be seen that the indexed set-based encoding length of $\mathcal{S}$ is $\sum_{i=1}^{k}|P_i|$, which is much smaller than that of binary encoding (i.e. $|\mathcal{I}| \times k$). The comparison results between these two representation scheme shown in the experimental section will demonstrate the effectiveness of the proposed indexed set-based representation scheme used in ISR-MOEA.

3.2. The overall procedure of ISR-MOEA

The proposed algorithm ISR-MOEA is implemented by using the proposed indexed set-based representation scheme and the framework of MOEA/D (Zhang and Li, 2007) to guarantee a good trade-off between the convergence and diversity of populations during evolution. Let pop be the population size, then the weight vectors $\lambda = \{\lambda_1, \lambda_2, \ldots, \lambda_{pop}\}$ are a set of weight vectors evenly spread on $\lambda_i^1 + \lambda_i^2 = 1$, where $\lambda_i = \langle \lambda_i^1, \lambda_i^2 \rangle$, $\lambda_i^1, \lambda_i^2 \in [0, 1]$ and $1 \leq i \leq pop$. The weight vectors λ_i are used to decompose the multi-objective pattern mining problem into pop single-objective subproblems. In ISR-MOEA, the proposed index-based representation scheme is used as the individual encoding, which have been introduced in the above subsection.

Algorithm 1 presents the general framework of ISR-MOEA, including three steps: the population initialization, population evolution and population selection. In the first step, the proposed initialization strategy is suggested to obtain the initial population (see Section 3.3). Then, the reference point z^* is obtained by computing the best values of *utility* and *coverage* found in the initial population. For each weight vector λ_i ($1 \leq i \leq pop$), the Euclidean distances between any individual ind in current population P and the weight vector λ_i are calculated. Then, the set of ns individuals (denoted as N_i) from P with the nearest Euclidean distance to λ_i are chosen as the neighbours of λ_i. In the second step, for each individual p_j in P ($1 \leq j \leq pop$), one individual ind is randomly selected from N_j. The two individuals p_j and ind are used to generate the child chd by using the proposed crossover and mutation operators (see Section 3.4). If the Tchebycheff value (Miettinen, 1998) of chd is better than an individual ind' in N_j, then replace ind' with chd and update reference point z^*. In the last step, the algorithm terminates when the maximum number of generations reaches, and the set of non-dominated solutions from the final population is obtained by fast non-dominated sorting strategy (Zhang et al., 2015a).

In summary, compared with MOEA/D algorithms (Zhang and Li, 2007; Li and Zhang, 2009), an indexed set representation scheme is suggested to fast encode and decode the top-k pattern set. Based on this representation, we propose an effective multi-objective evolutionary algorithm ISR-MOEA based on the framework of MOEA/D for mining diversified top-k high utility patterns, where a novel population initialization strategy and novel evolutionary operators are proposed for speeding up the convergence of populations. It is worth noting that the proposed algorithm ISR-MOEA just adopts the framework of MOEA/D, and other frameworks such as NSGA-II (Deb et al., 2002) and SPEA-II (Zitzler et al., 2001) can be easily extended with the proposed individual representation, population initialization strategy and evolutionary operators. In the following, we will illustrate the proposed population initialization strategy and evolutionary operators such as crossover and mutation in detail.

3.3. Population initialization strategy

The population initialization strategy for evolutionary algorithms has been studied extensively for single-objective problems and a recent survey could be found in Kazimipour et al. (2014). However, there were only a few studies that could be found on initialization techniques for multi-objective evolutionary algorithms (MOEAs). To be specific, Hernandezdiaz et al. (2008) proposed a population initialization strategy by using gradient-based information for NSGA-II (Deb et al., 2002) and the quality was assessed on the benchmark family ZDT. Friedrich and Wagner (2015) presented a computational study for seeding the initial population of MOEAs. They investigated the effect of two seeding techniques such as *CornersAndCentre* and *LinearCombinations* for five MOEAs on four common families of benchmark functions(i.e. DTLZ, LZ09, WFG and ZDT). However, these initialization techniques were designed for continuous optimization problems, which is not applicable for the task of mining diversified top-k high utility patterns since that the proposed task is a combination optimization problem.

Algorithm 1 General Framework of ISR-MOEA

Input: The transaction database D, maximum number of generations $maxGen$, population size pop, weight vector $\{\lambda_1, \lambda_2, \ldots, \lambda_{pop}\}$, the size of neighbours ns, mutation probability p;
Output: The final set of non-dominated solutions FS.
Step1: Population initialization.
$P \leftarrow Initialization(D, pop)$// Initialize a population of solutions with the proposed initialization strategy;
$z^* \leftarrow Initialization(P)$//Initialize reference point;
$N = \{N_1, N_2, \ldots, N_{pop}\} \leftarrow$ get the ns individuals from P with the nearest Euclidean distance between an individual in P and the weight vector λ_i for $i = 1$ to pop;
Step2: Population evolution.
for $i = 1$ to $maxGen$ **do**
  for $j = 1$ to pop **do**
    $p_j \leftarrow$ Select one individual randomly from N_j;
    $chd \leftarrow$ Generate children by the proposed crossover and mutation operators;
    $N_j \leftarrow$ Update individuals in N_j;
    $z^* \leftarrow$ Update reference point z^*;
  end for
end for
Step3: Population selection.
$FS \leftarrow Selection(P)$;//Select the non-dominated solutions from the final population P by the fast non-dominated sorting strategy

Recently, we proposed a problem-specific initialization technique by using two kinds of patterns (i.e, meta-pattern and transaction pattern) in MOEA for mining frequent and high utility patterns (Zhang et al., 2018). Indeed, this problem-specific initialization in MOEAs is effective for the task of mining frequent and high utility patterns, however, is not effective for the proposed task of mining diversified top-k high utility patterns due to the complexity of the proposed task. To this end, we need to design new population techniques in MOEAs for the proposed task. In the following, we suggest a novel effective population initialization strategy in MOEAs for mining diversified top-k high utility patterns.

In the proposed initialization strategy, we use one kind of pattern named *meta-pattern*, which contains only one item, to initialize the population. Take the database shown in Table 1 as an example, the patterns $\{Jacket\}$, $\{Shoe\}$, $\{Belt\}$, $\{Hat\}$, $\{Watch\}$, $\{Suitcase\}$ and $\{Scarf\}$ are 7 different *meta-patterns*. These *meta-patterns* can be used to generate diversified and useful solutions. Let pop be the number of initial population and n be the number of distinct items in database ($n = |\mathcal{I}|$). The support and utility values for each meta-pattern $\mathcal{I}_i$ are calculated, denoted as $sup(\mathcal{I}_i)$ and $ruti(\mathcal{I}_i)$ respectively ($1 \leq i \leq |\mathcal{I}|$).

To generate each individual containing k patterns, two steps are required by using the information of support and utility separately. In the first step, m ($m > k$) meta-patterns are randomly selected from $\mathcal{I}$ according to their support probability. In the second step, k meta-patterns are selected from m meta-patterns according to their utility probability. In order to get a proper m value automatically, it is reasonable to keep the same selection ratio on the two steps, i.e., $\frac{n}{m} = \frac{m}{k}$, thus, $m = \sqrt{n \times k}$ ($n \gg k$). Finally, pop individuals are generated according to the above two steps. The comparison results in the experimental section will demonstrate the effectiveness of the proposed population initialization strategy used ISR-MOEA.

3.4. Evolutionary operators

It is worth noting that the traditional evolutionary operators such as crossover and mutation strategies with the constant encode length cannot be directly applied for the proposed indexed set based representation scheme, since that the encoding length of each individual varies. To overcome this problem, in the following we will propose two effective evolutionary operators (i.e. crossover strategy and mutation strategy) for the proposed indexed set-based representation scheme.

3.4.1. The crossover operator

Given two individuals $S^1 = \langle S_1^1, \Delta, S_2^1, \ldots, \Delta, S_k^1 \rangle$ and $S^2 = \langle S_1^2, \Delta, S_2^2, \ldots, \Delta, S_k^2 \rangle$, the crossover operator can be applied to any two sub-individuals S_i^1 and S_i^2 $(1 \leq i \leq k)$. Specifically, for each two sub-individuals S_i^1 and S_i^2 $(1 \leq i \leq k)$, we can randomly generate two crossover vectors C_i^1 and C_i^2, whose length are the same as that of S_i^1 and S_i^2 respectively. The crossover vector for each sub-individual C_i is denoted as

$$C_i = \langle c_1, c_2, \ldots, c_m \rangle \tag{5}$$

where $c_j \in \{0, 1\}$, $1 \leq j \leq m$ and m is the number of items in sub-individual S_i. If $c_j = 1$, the corresponding element (or item) in the sub-individual S_i will be removed and be added to another sub-individual; otherwise, the corresponding element keeps unchanged. If there exists same index in both S_i^1 and S_i^2, it is not necessary to change this index in further crossover procedure, thus, the corresponding element in crossover vector is set to 0.

As shown in Fig. 2, there are two sub-individuals $S_i^1 = \{1, 2, 3, 7\}$ and $S_i^2 = \{2, 4, 5, 6, 7\}$. Then two corresponding crossover vectors C_i^1 and C_i^2 are generated randomly, where $C_i^1 = \langle 1, 0, 1, 0 \rangle$ and $C_i^2 = \langle 0, 0, 1, 1, 0 \rangle$. According to the crossover vectors, the elements 1 and 3 in S_i^1 should be removed and will be added to S_i^2, while the elements 5 and 6 in S_i^2 should be removed and will be added to S_i^1. Finally, we can get two new sub-individuals $newS_i^1 = \{2, 5, 6, 7\}$ and $newS_i^2 = \{1, 2, 3, 4, 7\}$. In this running example, we can find that there exist two common indices '2' and '7' in both S_i^1 and S_i^2, thus the corresponding element of indices '2' and '7' in the generated crossover vector should be set to 0.

3.4.2. The mutation operator

Given one individual $S = \langle S_1, \Delta, S_2, \ldots, \Delta, S_k \rangle$ for mutation with a probability of p. If S is selected for mutation, for each sub-individual S_i $(1 \leq i \leq k)$ in S, we can randomly generate a mutation vector M whose length is n (the number of distinct items in database), which is defined as

$$M_i = \langle m_1, m_2, \ldots, m_n \rangle \tag{6}$$

where $m_j \in \{0, 1\}$ and $1 \leq j \leq n$. $m_i = 1$ means that i-element should be changed in S_i. Specifically, if i-element is in S_i, then i-element should be removed from S_i; otherwise, the i-element will be added to S_i.

For example, as shown in Fig. 3, there exist two sub-individuals $S_i^1 = \{2, 5, 6, 7\}$ and $S_i^2 = \{1, 2, 3, 4, 7\}$. The two corresponding mutation vectors M_i^1 and M_i^2 are randomly generated, suppose $M_i^1 = \{1, 1, 0, 1, 0, 0, 0\}$ and $M_i^2 = \{1, 0, 0, 0, 1, 1, 0\}$. According to the definition of mutation vector, the elements 1, 2, 4 should be changed in S_i^1 and the elements 1, 5, 6 should be changed in S_i^2. To be specific, for S_i^1, the element 2 is contained in S_i^1, thus 2 should be removed from S_i^1. The elements 1, 4 are not contained in S_i^1, 1 and 4 should be added into S_i^1. Similarly for S_i^2, the element 1 should be removed and the elements 5, 6 will be added into S_i^2. Finally, we can get two new sub-individuals $newS_i^1 = \{1, 4, 5, 6, 7\}$ and $newS_i^2 = \{2, 3, 4, 5, 6, 7\}$.

4. Experimental results

In this section, we first introduce baselines and datasets. Then we compare ISR-MOEA with several baselines, and give a detailed analysis about the experiments.

4.1. Experimental settings

4.1.1. Comparison methods

The proposed multi-objective evolutionary algorithm ISR-MOEA is compared with several state-of-the-art algorithms and its variants.

- TKO: TKO (Tseng et al., 2016) is one of state-of-the-art algorithms for mining top-k high utility patterns, where only *utility* is used as the interestingness measure.
- TKO-Greedy: In order to adapt TKO for mining diversified top-k high utility patterns, we first use TKO for mining top-2k high utility patterns (denoted as $\mathcal{P}$), from which k patterns are then selected by greedy strategy on coverage (named TKO-Greedy). Initially, suppose pattern set $S = \Phi$. At each iteration, it adds a new pattern P from $\mathcal{P}$ into S if P maximizes the increment on *coverage*. Then P is removed form $\mathcal{P}$. This process terminates until k patterns are selected.
- PSO-Miner: Based on binary representation scheme, a single-objective evolutionary algorithm PSO (Liu et al., 2007) can be easily extended (named PSO-Miner) for mining diversified top-k high utility patterns, where the fitness function is shown in Eq. (1) and a parameter α should be specified by users.
- BGSA-Miner: Based on binary representation scheme, a single-objective binary gravitational search algorithm (Rashedi et al., 2010) can be easily extended (named BGSA-Miner) for mining diversified top-k high utility patterns, where the fitness function is shown in Eq. (1) and a parameter α should be specified by users. For solving continuous problems by using gravitational search algorithm, interested readers can refer to recent work (Pelusi et al., 2018a, b).
- SSDP-Miner: SSDP is a single-objective evolutionary approach for mining top-k discriminative patterns (Lucas et al., 2017). This algorithm can be extended for mining diversified top-k high utility patterns (named SSDP-Miner) by replacing the fitness function in SSDP as $(1 - \alpha) \cdot uti(S) + \alpha \cdot cov(S)$ for updating top-k patterns, where a parameter α should be specified by users.
- ISR-NSGA2: Based on the proposed indexed-set representation scheme and the proposed initialization strategy, the framework of NSGA-II (Deb et al., 2002) is extended for mining diversified top-k high utility patterns (named ISR-NSGA2).
- ISR-SPEA2: Based on the proposed indexed-set representation scheme and the proposed initialization strategy, the framework of SPEA-II (Zitzler et al., 2001) is extended for mining diversified top-k high utility patterns (named ISR-SPEA2).
- ISR-MOEA(Binary): In order to show the effectiveness of the proposed index based representation used in ISR-MOEA, we compare ISR-MOEA with a variant of ISR-MOEA, named ISR-MOEA(Binary), where all components keep the same as ISR-MOEA except for individual representation by using binary representation.
- ISR-MOEA(Support), ISR-MOEA(Utility) and ISR-MOEA(Random): In order to show the effectiveness of the proposed initialization strategy used in ISR-MOEA, we compare ISR-MOEA with three variants of ISR-MOEA, namely ISR-MOEA(Support), ISR-MOEA(Utility) and ISR-MOEA(Random), where all components keep the same as ISR-MOEA except for the population initialization by only using support or utility or random strategy respectively. Specifically, in ISR-MOEA(Support), k meta-patterns composing of one individual are randomly selected from $\mathcal{I}$ (distinct items in database) according to their support probability. In ISR-MOEA(Utility), k meta-patterns composing of one individual are randomly selected from $\mathcal{I}$ according to their utility probability. In ISR-MOEA(Random), k meta-patterns composing of one individual are randomly selected from $\mathcal{I}$.

Note that both TKO and TKO-Greedy can return one solution (k-pattern set) in one run. When α is fixed, PSO-Miner, SSDP-Miner and BGSA-Miner can obtain one solution in one run. Different from these algorithms, the proposed algorithm ISR-MOEA can get multiple solutions in only one run. These compared algorithms were implemented in Matlab language and the experiments were performed on a Windows 10 laptop with quad core Intel i5-7500 @3.4G HZ processor and 16 GB of main memory. For each baseline algorithm, we adopt the parameters suggested in their papers. For ISR-MOEA, the population size is set to 100, the weight vectors λ is set to

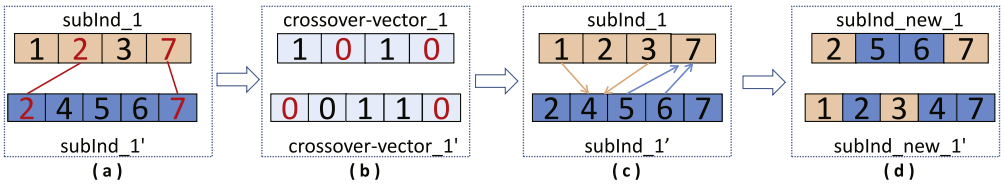


Fig. 2. The procedure of the proposed crossover operator.

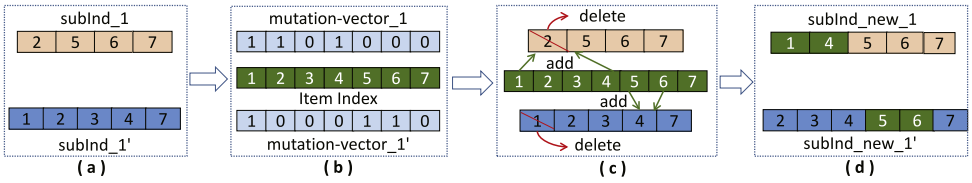


Fig. 3. The procedure of the proposed mutation operator.

Table 2
The 8 datasets with different characteristic.

Datasets	#Tran.	#Items	AveLen.	MaxLen.	Type
Chess	3,196	76	37	37	Dense
Mushroom	8,124	120	23	23	Dense
Connect_50%	33,779	129	43	43	Dense
Accidents_10%	34,018	469	34	46	Dense
Powerc_10%	104,000	133	7	7	Sparse
OnlineRetail_10%	54,191	2585	4.4	8	Sparse
D200kN3k	200,000	3000	11	36	Sparse
D500kN3k	300,000	3000	11	40	Sparse

$\{\langle\frac{0}{99},\frac{99}{99}\rangle,\langle\frac{1}{99},\frac{98}{99}\rangle,\langle\frac{2}{99},\frac{97}{99}\rangle,\ldots,\langle\frac{98}{99},\frac{1}{99}\rangle,\langle\frac{99}{99},\frac{0}{99}\rangle\}$, the maximal generation is set to 300, the size of neighbours is set to 10 and the mutation probability p is set to 0.15. In this paper, k is set to 5, and 30 independent runs are performed for each dataset for PSO-Miner, SSDP-Miner, BGSA-Miner, ISR-NSGA2, ISR-SPEA2 and ISR-MOEA.

4.1.2. Datasets description

In our experiments, 6 real databases with different characteristics are adopted, namely *Chess, Mushroom, Connect, Accidents, Powerc and OnlineRetail*. These datasets are popularly used in high utility pattern mining and can be downloaded from SPMF Repository (Fournier-Viger et al., 2014). In addition, 2 large and sparse synthetic datasets generated by IBM synthetic data generator (Agrawal and Srikant, 1995) are adopted, namely *D200kN3k,D300kN3k*. Table 2 summarizes the characteristics of these datasets. Note that the first 50% of Connect, 10% of Accidents, 10% of Powerc, 10% of OnlineRetail are adopted for comparison, since that the compared algorithms cannot run on these datasets if all transactions are used. In order to get the corresponding weights for these transactional databases, a similar method like the previous work (Zhang et al., 2016; Liu and Qu, 2012; Ahmed et al., 2009; Zhang et al., 2018) is adopted to generate the weights for items randomly ranging from 1 to 1000.

4.2. Experimental results

4.2.1. Comparison results

In order to show the effectiveness of the proposed ISR-MOEA, we compare ISR-MOEA with seven baselines: TKO, TKO-Greedy, PSO-Miner, SSDP-Miner, BGSA-Miner, ISR-NSGA2 and ISR-SPEA2. TKO and TKO-Greedy can only get one solution. Note that a weight parameter $\alpha \in [0,1]$ is used in PSO-Miner, SSDP-Miner and BGSA-Miner by optimizing the weighted sum of utility and coverage (see Eq. (1)). PSO-Miner, SSDP-Miner and BGSA-Miner can obtain one solution in one run while α is fixed. Different α value used in PSO-Miner, SSDP-Miner and BGSA-Miner may get different optimal solution. In order to get a set of solutions, different α are used in PSO-Miner, SSDP-Miner and BGSA-Miner. Specifically, we generate a number of α evenly sampled in [0,1] to get a set of solutions. To make a fair comparison, the number of α is equal to the size of non-dominated solutions obtained by the proposed ISR-MOEA.

Fig. 4 presents the solutions plots of different compared algorithms on the eight datasets. Note that the solutions plots of ISR-NSGA2 and ISR-SPEA2 very similar with ISR-MOEA (see Table 3), thus, we do not display the solutions plots of ISR-NSGA2 and ISR-SPEA2 in order to make the figure more succinct. From this figure, we can find that the final solutions obtained by PSO-Miner, SSDP-Miner and BGSA-Miner are almost dominated by those of ISR-MOEA on the eight datasets, which demonstrates that the proposed multi-objective evolutionary algorithm ISR-MOEA used for mining diversified top-k high utility patterns is better than the single objective evolutionary algorithms PSO-Miner, SSDP-Miner and BGSA-Miner. We can also find that TKO get the best utility value but the worst coverage value since TKO does not consider coverage at all. The coverage value of TKO-Greedy is slightly increased at the expense of utility compared to TKO. Compared with TKO and TKO-Greedy, the proposed ISR-MOEA can provide users with much more tradeoff solutions for selection.

In order to further show the difference, Table 3 gives the hypervolume (HV) values[1] of compared algorithms on the eight datasets. The Wilcoxon rank sum test at a significance level of 0.05 is adopted here to evaluate the statistical performance difference of PSO-Miner, SSDP-Miner, BGSA-Miner, ISR-NSGA2 and ISR-SPEA2, in which the symbols '+', '−' and '≈' represent that the result is significantly better, significantly worse and statistically similar to that obtained by ISR-MOEA, respectively. From the table, we can find that the HV obtained by using multi-objective evolutionary algorithms are all better than single-objective evolutionary algorithms or traditional algorithms such as TKO and TKO-Greedy on the eight datasets, which demonstrated the effectiveness of the proposed strategies such as initialization and evolutionary operators. However, the proposed algorithm ISR-MOEA by using the framework of MOEA/D is slightly better by using the framework of NSGA-II or SPEA-II, thus, MOEA/D is adopted as the basic framework in the proposed algorithm.

4.2.2. Effectiveness of the proposed strategies in ISR-MOEA

In this subsection, we will present the effectiveness of the proposed indexed set representation scheme and the proposed initialization strategy used in ISR-MOEA. For conciseness, we only report the experimental

[1] Hypervolume is a well-known performance metric for multi-objective optimization problems, which is used to measure the convergence and distribution (i.e. area value) of solutions in objective space. The larger values of hypervolume indicate better solutions (Zitzler and Thiele, 1998).

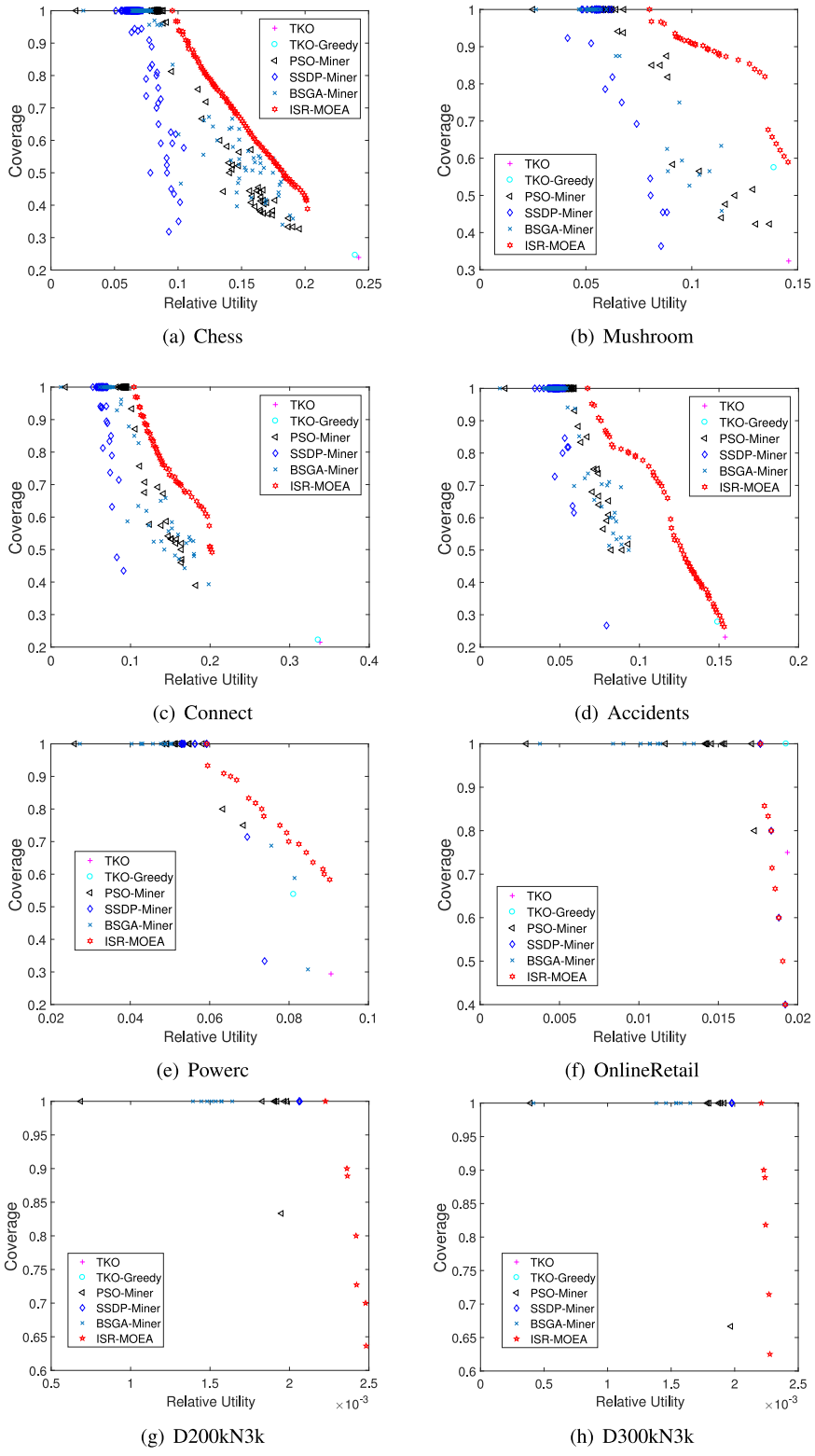


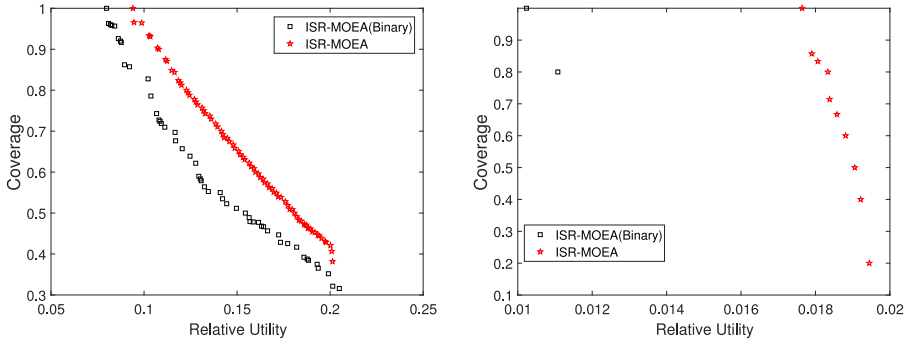
Fig. 4. The plots of obtained solutions in objective space for compared algorithms on the eight datasets.

results on two representative datasets (i.e. one dense dataset Chess and one sparse dataset OnlineRetail) and the proposed algorithm on the remaining six datasets achieve the similar results.

Fig. 5 presents the non-dominated solutions obtained by ISR-MOEA with index-based representation (denoted as ISR-MOEA) and ISR-MOEA with binary representation (denoted as ISR-MOEA(Binary)) on the datasets Chess and OnlineRetail. The numbers of non-dominated solutions on datasets Chess and OnlineRetail for ISR-MOEA are 78 and 7 respectively, while those for ISR-MOEA(Binary) are 51 and 2 respectively. In addition, we can find that the non-dominated solutions generated by ISR-MOEA(Binary) are almost dominated by ISR-MOEA, which demonstrates the effectiveness of the proposed indexed set-based individual representation.

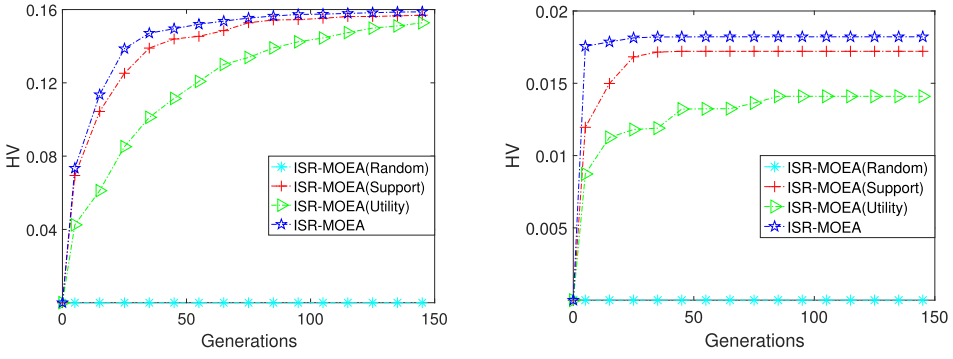
Table 3
The comparison results of different compared algorithms on the 8 datasets in terms of performance metric hypervolume, where symbols '+', '−' and '≈' indicate that the performance is significantly better, significantly worse and statistically similar to that of ISR-MOEA, respectively. Note that TKO and TKO-Greedy cannot run on the large datasets D200kN3k and D300kN3k in the current experimental setup.

Algorithms	TKO	TKO-Greedy	PSO-Miner	SSDP-Miner	BGSA-Miner	ISR-SPEA2	ISR-NSGA2	ISR-MOEA
Chess	0.0578	0.0589	$0.1479^{-} \pm 0.0017$	$0.0953^{-} \pm 0.0031$	$0.1548^{-} \pm 0.0037$	$0.1608^{-} \pm 0.0014$	$0.1578^{-} \pm 0.0009$	**0.1650** ± 0.0004
Mushroom	0.0472	0.0798	$0.1094^{-} \pm 0.0017$	$0.0771^{-} \pm 0.0025$	$0.1010^{-} \pm 0.0039$	$0.1301^{-} \pm 0.0011$	$0.1318^{\approx} \pm 0.0010$	**0.1325** ± 0.0006
Connect	0.0727	0.0746	$0.1483^{-} \pm 0.0025$	$0.0861^{-} \pm 0.0026$	$0.1517^{-} \pm 0.0030$	$0.1721^{\approx} \pm 0.0039$	$0.1705^{\approx} \pm 0.0041$	**0.1722** ± 0.0009
Accidents	0.0355	0.0414	$0.0825^{-} \pm 0.0016$	$0.0590^{-} \pm 0.0031$	$0.0848^{-} \pm 0.0025$	$0.1088^{-} \pm 0.0019$	$0.1090^{-} \pm 0.0014$	**0.1186** ± 0.0008
Powerc	0.0266	0.0437	$0.0743^{-} \pm 0.0038$	$0.0675^{-} \pm 0.0037$	$0.0694^{-} \pm 0.0080$	$0.0829^{-} \pm 0.0002$	$0.0835^{\approx} \pm 0.0008$	**0.0836** ± 0.0005
OnlineRetail	0.0145	**0.0193**	$0.0169^{-} \pm 0.0007$	$0.0186^{-} \pm 0.0000$	$0.0127^{-} \pm 0.0013$	$0.0185^{-} \pm 0.0002$	$0.0186^{\approx} \pm 0.0002$	0.0187 ± 0.0000
D200kN3k	–	–	$0.0020^{-} \pm 0.0001$	$0.0021^{-} \pm 0.0000$	$0.0017^{-} \pm 0.0001$	$0.0022^{\approx} \pm 0.0001$	$0.0022^{\approx} \pm 0.0001$	**0.0023** ± 0.0001
D300kN3k	–	–	$0.0020^{-} \pm 0.0001$	$0.0020^{-} \pm 0.0001$	$0.0016^{-} \pm 0.0001$	$0.0022^{\approx} \pm 0.0001$	$0.0023^{\approx} \pm 0.0001$	**0.0023** ± 0.0001
+/−/≈	–	–	0/8/0	0/8/0	0/8/0	0/5/3	0/2/6	



(a) Chess (b) OnlineRetail

Fig. 5. The non-dominated solutions obtained by ISR-MOEA with index-based representation and ISR-MOEA with binary representation (ISR-MOEA-Binary) on Chess and OnlineRetail.



(a) Chess (b) OnlineRetail

Fig. 6. The hypervolume (HV) of non-dominated solutions obtained by ISR-MOEA and its variants with different generations on datasets Chess and OnlineRetail.

Table 4
The running time (seconds) between ISR-MOEA with binary representation (denoted as ISR-MOEA-Binary) and ISR-MOEA with index-based representation (denoted as ISR-MOEA) on the eight datasets, averaging on 30 runs.

Datasets	#Tran.	#Items	Type	Time(ISR-MOEA-Binary)	Time(ISR-MOEA)
Chess	3,196	76	dense	51.43 ± 3.25	**45.27** ± **0.65** s
Mushroom	8,124	120	dense	142.84 ± 17.09	**79.41** ± **1.19** s
Connect_50%	33,779	129	dense	1706.80 ± 124.72	**1373.40** ± **24.16** s
Accidents_10%	34,018	469	dense	774.29 ± 83.68	**502.41** ± **5.80** s
Powerc_10%	104,000	133	sparse	930.52 ± 112.42	**727.50** ± **80.97** s
OnlineRetail_10%	54,191	2585	sparse	2472.10 ± 4.68	**96.51** ± **2.28** s
D200kN3k	200,000	3000	sparse	9677.60 ± 46.06	**633.19** ± **76.13** s
D300kN3k	300,000	3000	sparse	15930.10 ± 106.81	**947.19** ± **114.15** s

Fig. 6 shows the hyper-volume values of non-dominated solutions obtained by ISR-MOEA and the baselines with different generations on datasets Chess and OnlineRetail. The baselines ISR-MOEA(Support), ISR-MOEA(Utility) and ISR-MOEA(Random) are the three variants of ISR-MOEA, where all components keep the same as ISR-MOEA except for the population initialization by only using support or utility or random strategy respectively. From this figure, we can observe that ISR-MOEA can converge very quickly within 100 generations on both

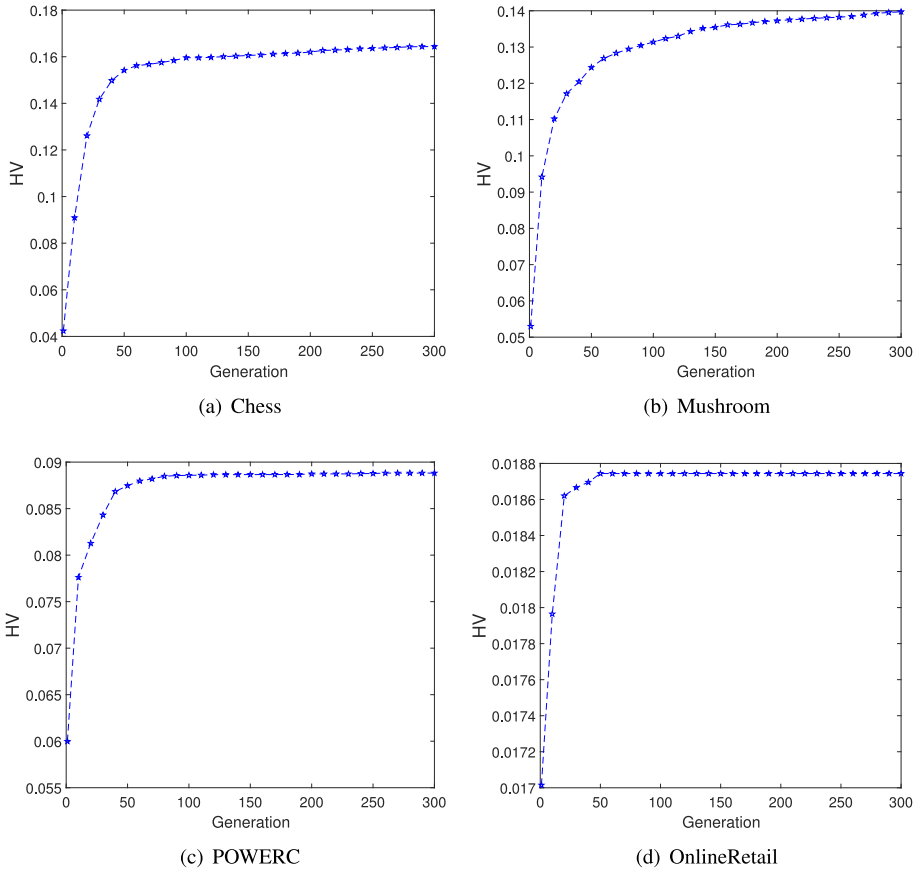


Fig. 7. The hypervolume (HV) of non-dominated solutions obtained by ISR-MOEA on four datasets Chess, Mushroom, POWERC and OnlineRetail with different maximum number of generations $maxGen$ and the mutation probability p is fixed to 0.15.

datasets Chess and OnlineRetail. We can also observe that ISR-MOEA with the proposed population initialization strategy by using both support and utility information can get better performance than using only support (ISR-MOEA(Support)) or utility (ISR-MOEA(Utility)). Note that random initialization used in ISR-MOEA cannot coverage at all due to that many generated solutions do not appear in transaction database.

4.2.3. Efficiency of ISR-MOEA

Let m be the number of transactions in database $\mathcal{D}$ and n be the number of distinct items in $\mathcal{D}$, k be the number of patterns in a pattern set, $maxGen$ be the maximum number of generations, pop be the size of population and $avgLen$ be the average length of patterns. The major computational complexity of ISR-MOEA is taken for the calculation of objective function values of individuals in the step of population evolution, thus the computational complexity of this step is $O(maxGen * pop * k * m * avgLen)$. For ISR-MOEA with binary representation, the computational complexity is $O(maxGen * pop * k * m * n)$, and the average length of pattern $avgLen$ is usually much smaller than the number of distinct items n.

In this following, we will present the efficiency of ISR-MOEA for mining diversified top-k high utility patterns to verify the above analysis. Table 4 shows the comparison results between the running time of ISR-MOEA with binary representation (i.e. ISR-MOEA-Binary) and ISR-MOEA with indexed set-based representation (i.e. ISR-MOEA) on the eight datasets, averaging on 30 runs. From this table, we can observe that ISR-MOEA with indexed set-based representation is faster than ISR-MOEA with binary representation on eight datasets, and the difference gap becomes larger when database becomes sparser. For example, the proposed ISR-MOEA runs 1 orders of magnitude faster than ISR-MOEA-Binary on large and sparse datasets OnlineRetail_10%, D200kN3k, and D300kN3k. Base on the above complexity analysis and experiments, we can make a conclusion that ISR-MOEA with the proposed index set-based representation can speed up the convergence of MOEAs especially on larger and sparser datasets.

4.2.4. Parameter analysis in ISR-MOEA

In the proposed algorithm ISR-MOEA, there are 5 initial parameters: population size pop, weight vector $\{\lambda_1, \lambda_2, \ldots, \lambda_{pop}\}$, the size of neighbours ns, maximum number of generations $maxGen$, mutation probability p. The first three parameters pop, $\{\lambda_1, \lambda_2, \ldots, \lambda_{pop}\}$ and ns are usually the same in MOEA/D algorithms for different applications, thus, we use the recommended parameters in MOEA/D algorithms directly. For the last two parameters $maxGen$ and p, there may be different optimal values for different applications, thus, we adjust one parameter while fixing all the others to show the performance changes, and then select the best parameter combinations. For conciseness, we only report the experimental results on four representative datasets (i.e. two dense datasets Chess and Mushroom, two sparse datasets POWERC and OnlineRetail) and the proposed algorithm on the remaining four datasets achieve the similar results.

Fig. 7 shows the hyper-volume values of non-dominated solutions obtained by ISR-MOEA on four datasets Chess, Mushroom, POWERC and OnlineRetail with different maximum number of generations $maxGen$ and the mutation probability p is fixed to 0.15. From this figure, we can observe that the proposed algorithm can converge within 100 generations for sparse datasets POWERC and OnlineRetail, and within 300 generations for dense datasets Chess and Mushroom. Thus, in this paper, the $maxGen$ is fixed to 300. Fig. 8 shows the hyper-volume values of non-dominated solutions obtained by ISR-MOEA on datasets Chess, Mushroom, POWERC and OnlineRetail with different mutation probability p and the $maxGen$ is fixed to 300. From this figure, we can see that the proposed algorithm can get the best performance with the p

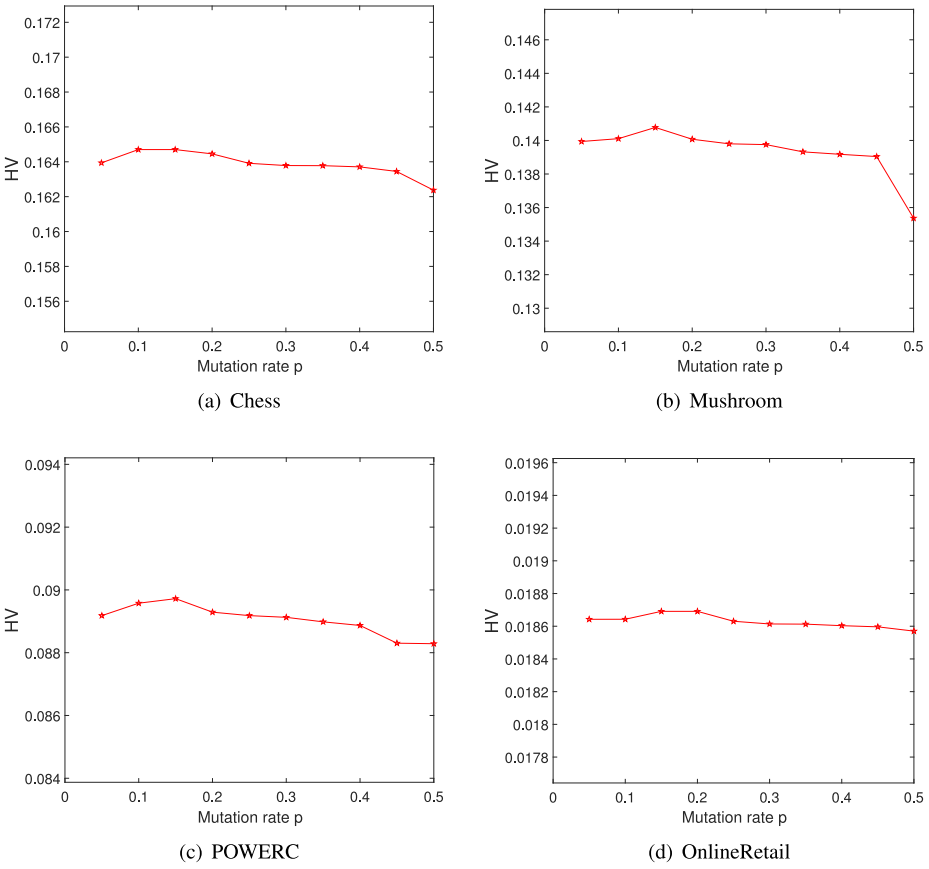


Fig. 8. The hypervolume (HV) of non-dominated solutions obtained by ISR-MOEA on four datasets Chess, Mushroom, POWERC and OnlineRetail with different mutation probability p and the $maxGen$ is fixed to 300.

ranging from 0.1 to 0.2 on these datasets. Thus, the mutation probability p is set to 0.15 in this paper.

5. Conclusions and future work

In this paper, we have introduced a metric *coverage* for measuring the diversity of top-k high utility patterns and formulated the novel task of mining diversified top-k high utility patterns as a 2-objective optimization problem. To solve this problem, we proposed an effective multi-objective evolutionary algorithm named ISR-MOEA to obtain many diversified top-k high utility patterns. In ISR-MOEA, a novel indexed set based representation and the corresponding crossover and mutation strategies were proposed to guide the direction of evolutions and speed up the convergence of populations. In addition, a novel initialization strategy by using the support and utility information was proposed to guarantee that all generated solutions are useful and hold a good diversity for recommendation. It is worth noting that the proposed algorithm can simultaneously provide multiple recommendations for decision makers in only one run. The experimental results on several real and synthetic datasets demonstrated the effectiveness of the proposed indexed set representation and the proposed initialization strategy used in ISR-MOEA. Note that the concept of coverage is useful in many pattern mining applications, which can be extended to sequential pattern mining (Agrawal and Srikant, 1995; Fradkin and Mörchen, 2015) and graph mining (Washio and Motoda, 2003) in the future work.

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